

Recommendation Scores

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check a boundary for Weighted scores and constraints. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Weighted scores and constraints. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate scores with $2 * \text{fun} + \text{learning}$ and choose.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Score = $3 \times \text{genreMatch} + 2 \times \text{short} + \text{illustrated}$. Compute $A=[1,0,1]$, $B=[0,1,1]$, $C=[1,1,0]$. A reader now requires illustrations. Which book wins after applying that constraint?

6. Solve the given case

Score = 2 times genre match + length match. A matches both, B genre only, C length only.

Score and rank.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute scores and justify the ranking.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 23 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Publish the features, weights, raw scores, and reason for the top result.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Score= $3 \times \text{genreMatch} + 2 \times \text{short} + \text{illustrated}$. Compute $A=[1,0,1]$, $B=[0,1,1]$, $C=[1,1,0]$. A reader now requires illustrations. Which book wins after applying that constraint?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate a score

An eligible activity has fun 2 and learning 4. Use $2 * \text{fun} + \text{learning}$.

My reasoning and evidence

13. W2 Check a boundary

Which durations 19, 20, 21 qualify under at most 20?

My reasoning and evidence

14. W3 Repair selection

An ineligible item scores 15 and an eligible item scores 9. Which should be offered under the stated policy?

My reasoning and evidence

15. W4 Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate scores with $2 * \text{fun} + \text{learning}$ and choose.

My reasoning and evidence

16. W5 Test sensitivity

For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare weights (2,1) with (1,2).

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Recommendation Scores

A fictional book-activity chooser has a 14-minute limit.

Score = $2 \times \text{interest} + \text{practice}$. $A = (\text{interest}3, \text{practice}2, \text{time}14)$; $B = (4, 3, 15)$; $C = (2, 5, 12)$.

Require whole nonnegative feature values; time must be at most the limit. Highest score wins; tie by ID.

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A: $2 \times 3 + 2 = 8$. C: $2 \times 2 + 5 = 9$.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Why does B not become the winner when its score is largest?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Publish the recommender's constraints, score formula and tie rule; include component-level reasons in each suggestion.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Publish the recommender's constraints, score formula and tie rule; include component-level reasons in each suggestion.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-activity chooser has a 14-minute limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's score 11 cannot override its failed time requirement.

1. Choose
2. Explain exclusion
3. Eligibility first
4. Scores

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Explain exclusion B's score 11 cannot override its failed time requirement. A score combines preferences according to explicitly chosen weights; it does not erase hard constraints. Computing a high score for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and score components for explanations.

Practice 2: explain the deciding evidence

Changed-case practice: Change interest weight from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. A scores 11; C also scores 11; tie-by-ID chooses A.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A scores 11; C also scores 11; tie-by-ID chooses A. B remains too long; changing preferences changes only ranking of A and C.

Repair a missing building block

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank training neighbours. Short practice: Why do $k=1$ and $k=3$ disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
